EV Scooter Assembly Line Optimization

Lean Manufacturing + OEE + Bottleneck Analysis

Portfolio project based on synthetic manufacturing data

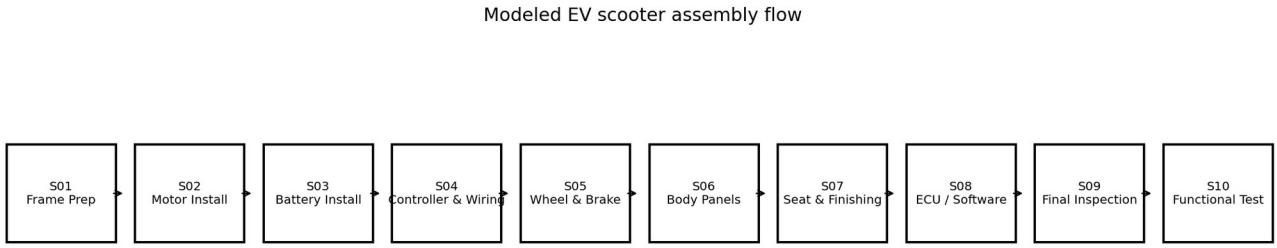


Figure 1. Modeled 10-station EV scooter assembly flow.

1. Executive Summary

Objective. Diagnose why a simulated EV scooter assembly line misses its production plan and define a Lean improvement roadmap that links cycle time, OEE, downtime, defects, and capacity.

Current state. Across 30 production days (600 station-shift records), the modeled final station processed 11,560 units against 15,502 planned units (74.6% plan attainment) and delivered 11,225 good units (97.1% good yield). The reference takt is 100.8 seconds; S05 Wheel & Brake averages 117.5 seconds, 16.7 seconds (16.6%) above takt.

Priority logic. S05 is treated as the primary cycle-time constraint. The largest loss buckets are material shortage and minor stops; the largest defect modes are fitment/alignment and wiring/connector. The improvement sequence therefore starts with S05 work-content/tooling, then S04 wiring/poka-yoke and material flow.

2. Project Scope and Dataset

Item	Value
Production horizon	30 production days
Station-shift records	600
Stations	10
Shifts	2 per day
Reference daily demand	500 units/day
Reference takt	100.8 sec/unit
Data status	Synthetic / simulated; not Ather factory data

Source alignment. The supplied Ather Operational Excellence JD emphasizes flow mapping, Muda/waste identification, ground-zero problem solving and visual tracking; the supplied Manufacturing Operations JD emphasizes process modification, PFMEA, poka-yoke, pilot production, plan-vs-actual, takt time and OEE. This project is intentionally structured around those capabilities.

3. Current-State Line Balance

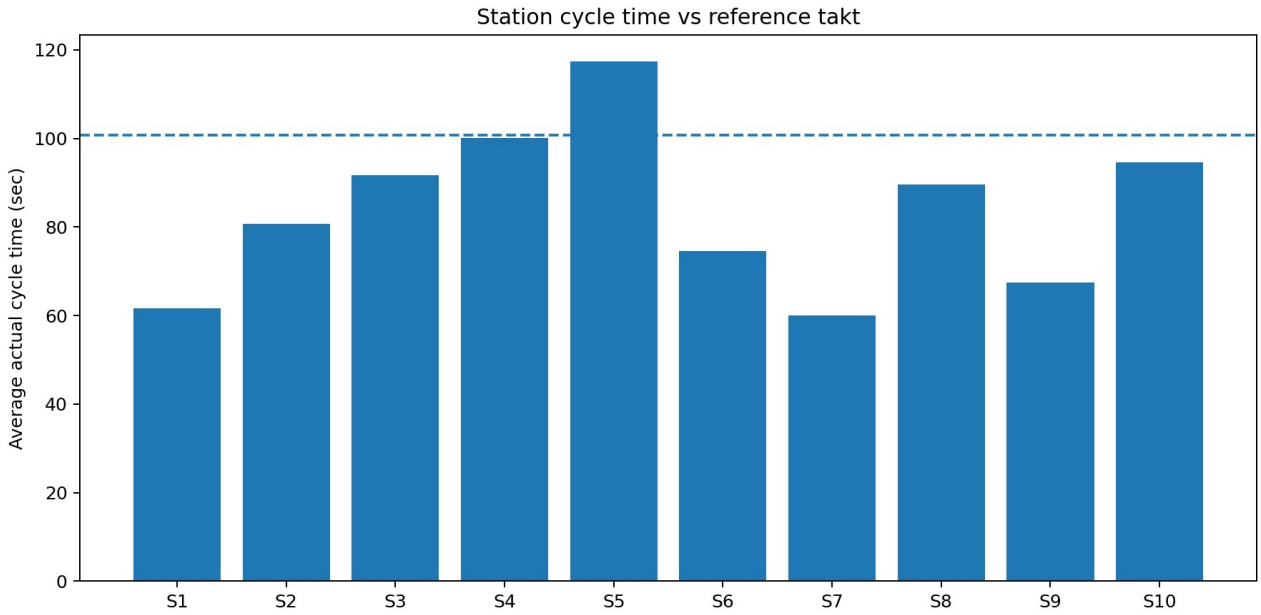


Figure 2. Average actual cycle time by station with a reference takt line at 100.8 seconds.

ID	Station	Ideal CT	Actual CT	Gap vs Takt	OEE	Plan Att.
S01	Frame Prep	60	61.6	-39.2	48.2%	76.9%
S02	Motor Install	78	80.8	-20.0	62.6%	76.8%
S03	Battery Install	88	91.7	-9.1	69.9%	77.0%
S04	Controller & Wiring	95	100.1	-0.7	73.7%	76.6%
S05	Wheel & Brake	110	117.5	+16.7	83.7%	73.9%
S06	Body Panels	72	74.6	-26.2	55.8%	75.0%
S07	Seat & Finishing	58	60.1	-40.7	45.6%	75.3%
S08	ECU / Software	85	89.6	-11.2	64.7%	74.6%
S09	Final Inspection	65	67.4	-33.4	50.1%	74.5%
S10	Functional Test	90	94.7	-6.1	69.0%	74.6%

Interpretation. S05 has the largest positive cycle-time gap (16.7 sec). Its OEE is not the lowest station OEE (83.7%), which illustrates why bottleneck identification should use both cycle-time and OEE/loss views rather than selecting the station with the lowest OEE automatically.

4. OEE Analysis

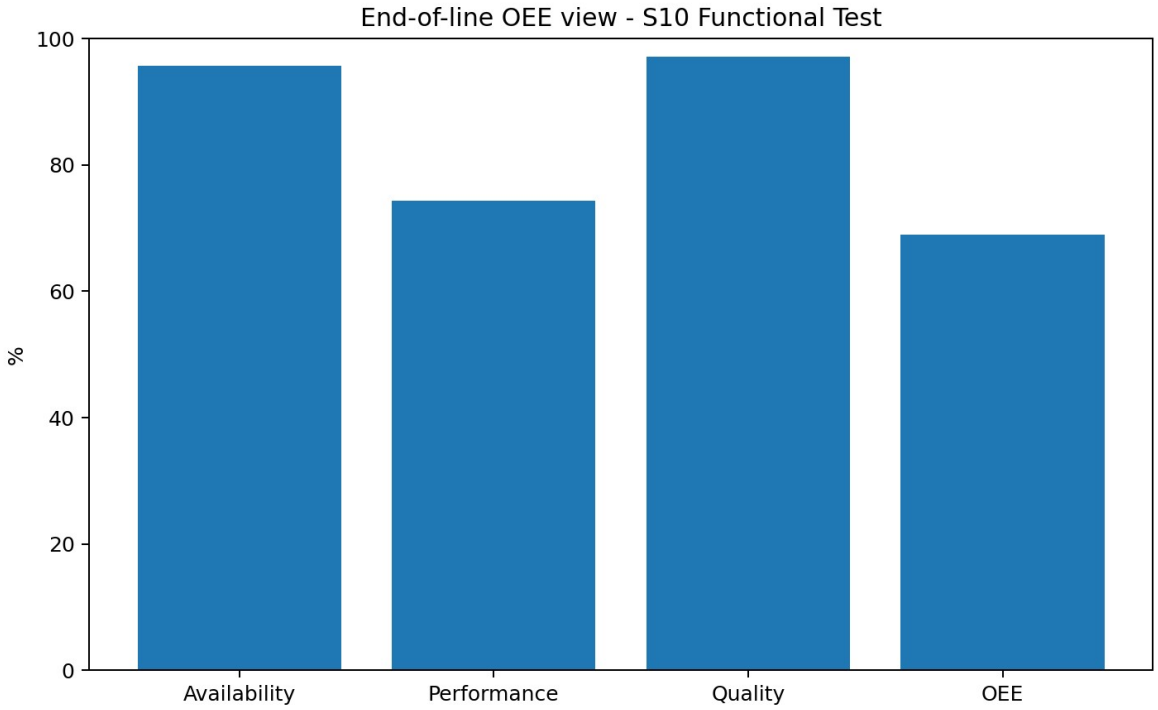


Figure 3. End-of-line OEE components for S10 Functional Test.

Metric	S10 / Line View
Availability	95.7%
Performance	74.3%
Quality	97.1%
Weighted OEE	69.0%

Reading the OEE. Availability is relatively high while performance is the largest component drag at the end-of-line station. This is consistent with a line that is not using the scheduled production window to its full theoretical rate; it should be read together with starvation, bottleneck and plan-attainment evidence.

5. Downtime and Quality Losses

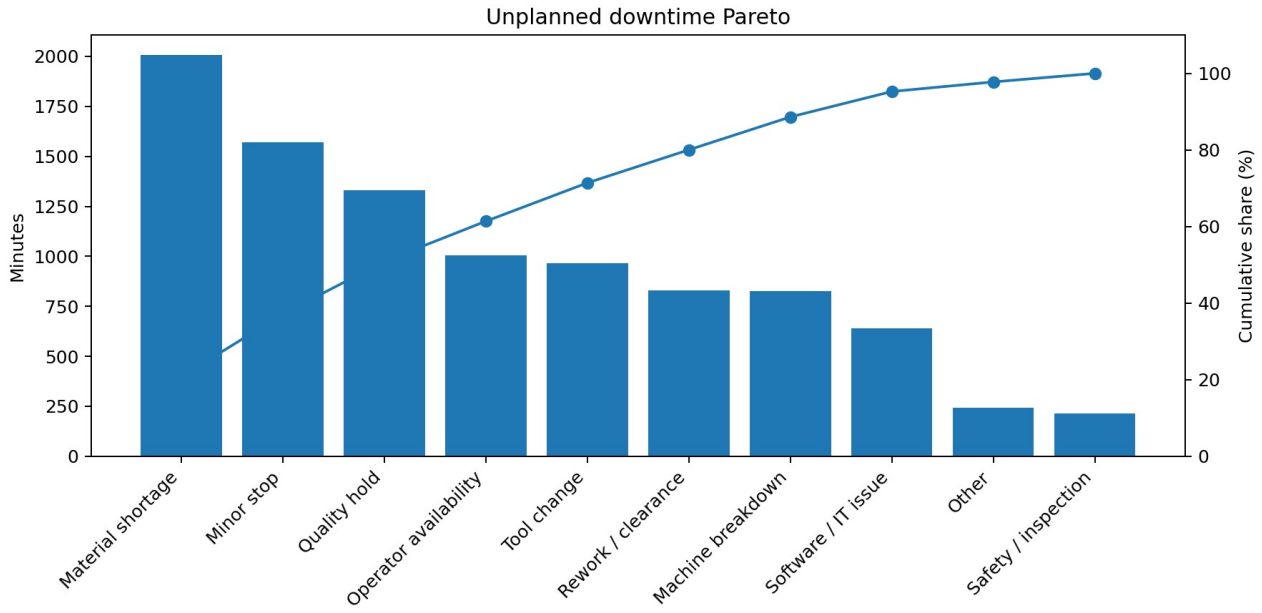


Figure 4. Unplanned downtime Pareto across all station-shift records.

Rank	Downtime reason	Minutes	Share
1	Material shortage	2,006	20.8%
2	Minor stop	1,572	16.3%
3	Quality hold	1,329	13.8%
4	Operator availability	1,006	10.4%
5	Tool change	965	10.0%
6	Rework / clearance	831	8.6%

Key signal. Material shortage is 20.8% of unplanned downtime and minor stops are 16.3%. This points to a combined material-availability and micro-stop problem rather than a single catastrophic machine failure.

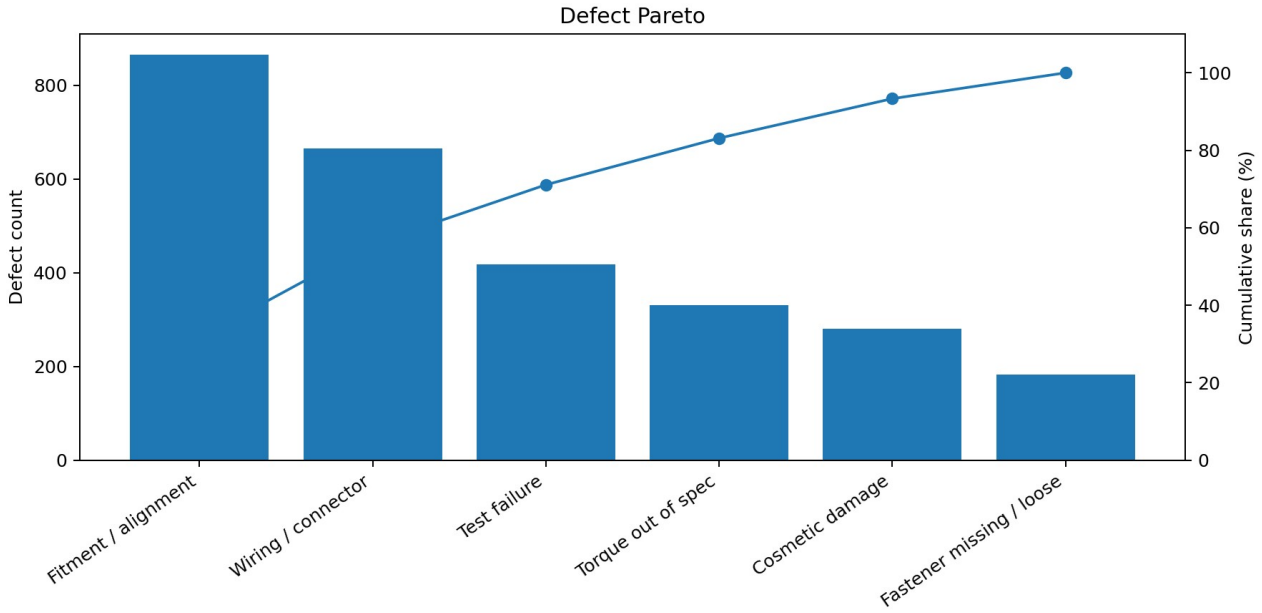


Figure 5. Defect Pareto across all station-shift records.

Rank	Defect mode	Count	Share
1	Fitment / alignment	866	31.6%
2	Wiring / connector	666	24.3%
3	Test failure	418	15.2%
4	Torque out of spec	330	12.0%
5	Cosmetic damage	280	10.2%
6	Fastener missing / loose	183	6.7%

6. Root Cause Analysis - S05

Problem statement. S05 Wheel & Brake averages 117.5 sec, exceeding the 100.8 sec reference takt by 16.7 sec.

Level	Logic	Evidence	Countermeasure
Why 1	Why is S05 a constraint?	Cycle time is above takt.	Time-study work content and rebalance.
Why 2	Why is cycle time high?	Dataset flags tooling variability and fitment/alignment as important loss signals.	Separate prep from critical-path work; improve fixture presentation.
Why 3	Why does variation add delay?	S05 carries tool-change and rework/clearance downtime.	Quick-change tooling; standardize torque / alignment sequence.
Why 4	Why do defects create extra work?	Fitment/alignment is the largest defect family.	Add poka-yoke, locators and first-piece checks.
Why 5	Why does it recur?	The dataset represents recurring station-level loss patterns.	Daily visual CT/OEE review with escalation trigger.

Important limitation. The last two “whys” are hypotheses to verify at the shop floor; the synthetic dataset supports the loss pattern, not a claim about an actual Ather process.

7. Illustrative PFMEA and Control Plan

The workbook contains a six-row illustrative PFMEA for S05 covering wheel alignment, torque, caliper position, fastener presence, tool change and rework clearance. A separate control-plan CSV turns those risks into measurable controls and reactions. These are portfolio artifacts, not audited factory documents.

Process	Failure mode	Effect	S	O	D	RPN
Wheel fitment	Misalignment	Fitment / rework	7	5	6	210
Brake torque	Out of spec	Safety / quality	9	4	5	180
Brake caliper	Position error	Drag / test failure	8	4	5	160
Fastener	Missing / loose	Rework / loose assembly	8	3	6	144
Tool change	Setup delay	Cycle-time loss	5	6	5	150
Rework clearance	Repeated adjustment	CT / WIP growth	6	6	5	180

8. Improvement Model

Stage 1 assumption. Reduce S05 cycle time from 117.5 sec to 99.5 sec and reduce S05 unplanned downtime by 25%. The modeled S05 capacity proxy rises from 11,888 to 14,182 units over the 30-day window, a theoretical increase of 19.3%.

Next constraint. At the current-state numbers, S04 Controller & Wiring becomes the next tight constraint with a capacity proxy of about 13,980 units over the 30-day window. This is why the project is framed as a sequence of bottleneck releases rather than a one-time fix.

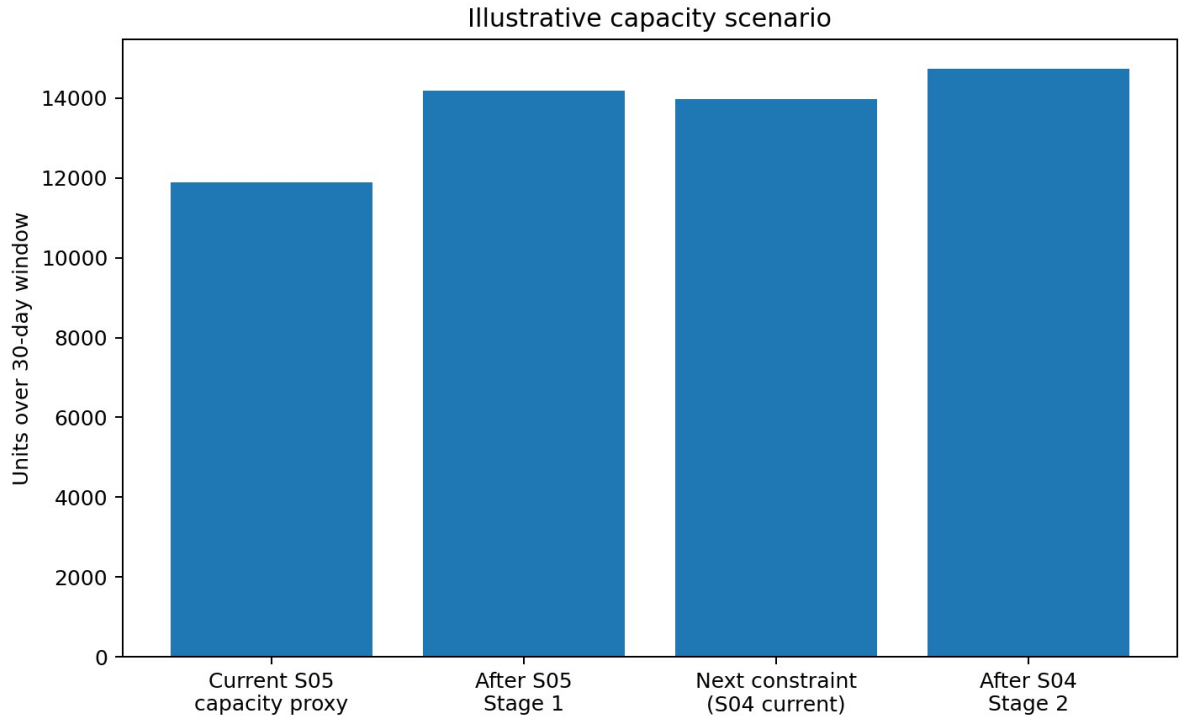


Figure 6. Illustrative capacity scenario; not a guaranteed production forecast.

Metric	Current	Illustrative target	Mechanism
S05 cycle time	117.5 sec	99.5 sec	Work balancing + fixture/tooling
S05 unplanned downtime	961 min	721 min	Quick change + preventive action
S03 material shortage DT	517 min	310 min	Kitting / supermarket / replenishment
S04 defects	461	323	Wiring standard work + poka-yoke

9. Daily Trend and Validation

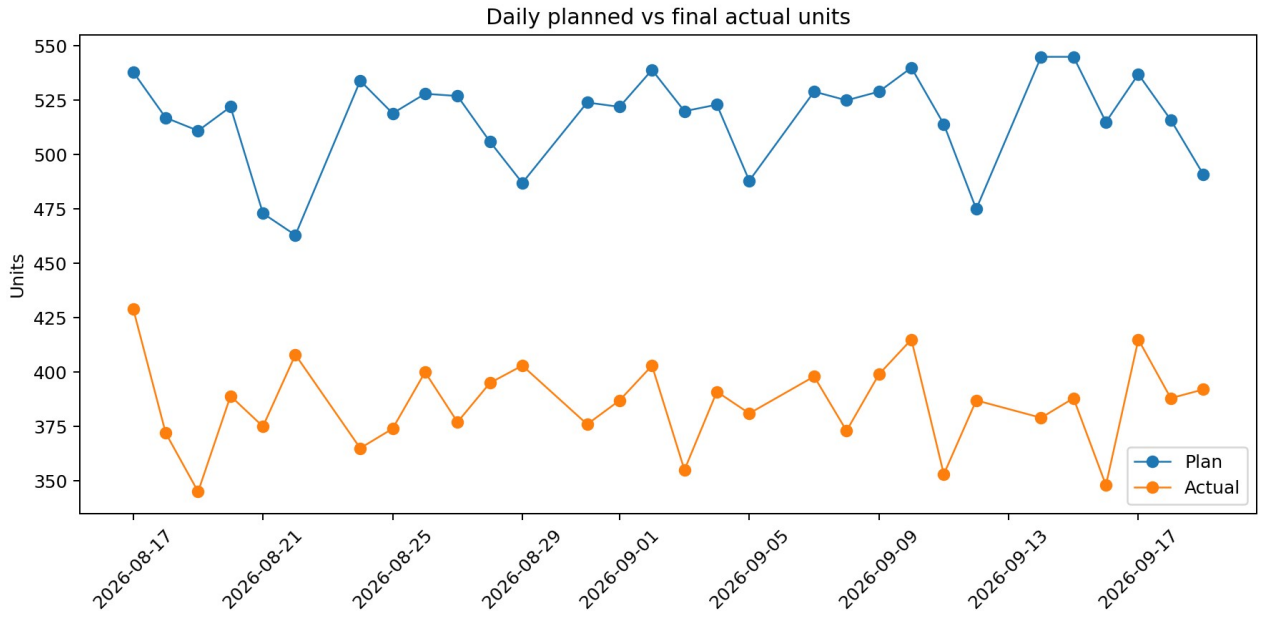


Figure 7. Daily plan vs final actual units.

Validation plan: run a pilot on S05, record a minimum two consecutive production shifts with a stable standard work method, verify the cycle-time distribution (not only the mean), monitor defects and rework, and then release the next improvement wave on S04. Use a visual daily management board for escalation and closure.

Date	Modeled condition
2026-08-24	Supplier delay
2026-09-03	Line trial / changeover-heavy day
2026-09-12	Software release / test instability

10. 30-Day Implementation Roadmap

When	Workstream	Deliverable	Owner
Week 1	Gemba + time study	S05 element breakdown, spaghetti diagram, CT distribution	IE / PM
Week 2	Quick wins	Fixture locator, torque sequence, kitting, quick-change standard	ME + Quality
Week 3	Pilot production	Run pilot, verify CT / OEE / defects, update PFMEA and control plan	Ops + Quality
Week 4	Control + scale	Visual dashboard, escalation rule, next constraint review (S04)	PM / Site Ops

11. Resume and Interview Translation

- Built a 10-station EV assembly-line simulation with 600 station-shift records and analyzed takt time, cycle time, OEE, downtime, defects and plan-vs-actual performance.

- Identified S05 Wheel & Brake as the primary cycle-time constraint at 117.5s vs 100.8s takt; designed a Lean bottleneck-release roadmap using time study, fixture/poka-yoke and quick-change concepts.
- Used Pareto + 5-Why + illustrative PFMEA/control plan to prioritize material shortage, fitment/alignment and wiring losses, with a formula-driven future-state capacity scenario.

12. Files and Reproducibility

The accompanying ZIP contains the final Excel workbook, original synthetic dataset workbook, CSV extract, Python rerun script, Power BI build specification, report DOCX/PDF, presentation PPTX, control-plan CSV and supporting charts.

Source note. Ather job descriptions supplied by the user in this conversation: Graduate Trainee - Operational Excellence (2 pages) and Program Manager Manufacturing Operations (2 pages). The project content is aligned to those supplied requirements. All manufacturing data in this project are simulated.